TaxGuard AI
COMPLETE AI AGENT ARCHITECTURE

AI AGENT ECOSYSTEM
Automated Intelligence. Human Oversight. Compliant Tax Operations.

Accuracy by Design | Human Oversight | Compliant & Secure | Scalable Operations

INPUT SOURCES
- **Client Portal** — Documents, tax information, questionnaires
- **Document Upload** — W-2, 1099, K-1, Statements, etc.
- **Email Intake** — Client communications
- **Integrations** — Banks, Payroll, Accounting Software (QuickBooks, etc.)
- **Government Portals** — IRS, State Tax Agencies (e-File, transcripts)
- **Manual Data Entry** — Forms and client interviews

EXTERNAL AI SERVICES
- **LLM (Large Language Model)** — Analysis, reasoning, Q&A
- **OCR / Document AI** — Extract data from tax documents
- **Speech-to-Text** — Transcribe client interviews
- **Vector Search** — Knowledge base and templates
- **AI Model Management** — Prompting, fine-tuning, guardrails

1 AI ORCHESTRATOR / SUPERVISOR AGENT
Coordinates all agents, manages workflow, prioritizes tasks, monitors progress, ensures compliance and human oversight.

- Task planning & routing
- Cross-agent coordination
- Exception handling
- Real-time monitoring
- Compliance enforcement

2 CLIENT AGENTS
Client Interaction & Support

- Client onboarding agent
- Document request agent
- Client communication agent
- Status update agent
- FAQ and guidance agent

3 TAX AGENTS
Tax Preparation & Analysis

- Data extraction agent (OCR/AI)
- Tax classification agent
- Calculation agent
- Form preparation agent
- Tax planning agent
- State tax agent

4 OPERATIONS AGENTS
Workflow & Service Management

- Task orchestration agent
- Scheduling agent
- Resource allocation agent
- Quality control agent
- Client follow-up agent
- Reporting and analytics agent

5 ACCOUNTING AGENTS
Financial Data & Reconciliation

- Chart of accounts agent
- Transaction categorization agent
- Bookkeeping agent
- Reconciliation agent
- Financial statement agent

6 COMPLIANCE AGENTS
Regulatory & Rules Enforcement

- IRS rule validation agent
- State tax compliance agent
- Document completeness agent
- Error and anomaly detection agent
- Regulatory update agent
- Risk assessment agent

7 SECURITY AGENTS
Data Protection & Access Control

- Identity verification agent
- Access control agent
- Data encryption agent
- Threat monitoring agent
- Audit log agent
- Compliance monitoring agent

8 HUMAN AUTHORIZATION LAYER
Critical decisions and final review by authorized personnel

- **Preparer** — Prepares returns, reviews AI work, ensures completeness
- **Reviewer** — Independent review, validates accuracy, checks compliance
- **CPA / EA** — Final approval, sign-off, e-filing authorization

Approval Controls
- Role-based approvals
- Maker-Checker workflow
- Comments and annotations
- Required for sensitive actions
- Full audit trail

9 AUTHORITATIVE RECORD
Finalized tax returns, supporting documentation and audit trail

- Immutable audit log
- Version controlled records
- Secure storage (encrypted)
- Long-term retention
- Search and retrieval

SYSTEM INTEGRATIONS
- **Accounting Software** — QuickBooks, Xero, Sage, etc.
- **Payment Processing** — Stripe, ACH, Card Payments
- **IRS Systems** — MeF, Transcripts, e-File (via partner)
- **State Tax Portals** — All U.S. States (via partner)
- **Email & Calendar** — Microsoft 365, Google Workspace
- **Document Storage** — Secure cloud storage (S3, GCS, etc.)

MONITORING & ANALYTICS
- **Agent Activity Dashboard** — Real-time status and performance
- **System Health Monitoring** — Uptime, errors, alerts
- **Audit & Compliance Reports** — Access logs, actions, approvals
- **Operational Analytics** — Throughput, cycle time, bottlenecks

INFRASTRUCTURE LAYER
Secure, Scalable and Compliant (e.g., AWS / Azure / GCP)

Compute	Database	File Storage	Security	Backup & DR	Monitoring	CI/CD
Containers / Kubernetes	PostgreSQL (Supabase)	Encrypted Storage	WAF, Firewall, DLP	Multi-Region	Logs, Metrics, Alerts	Automated Deployment

TaxGuard AI
COMPLETE AI AGENT ARCHITECTURE

AI-POWERED TAX OPERATING SYSTEM
61 Specialized Agents | Human Oversight | Compliant | Secure | Scalable

Automate Workflows | Increase Accuracy | Reduce Risk Risk | Improve Productivity | Better Client Experience

INPUT SOURCES

- **Client Portal** — Forms & Onboarding
- **Documents** — W-2, 1099, K-1, etc.
- **Email** — Client Communications
- **Accounting Software** — QuickBooks, Xero, etc.
- **Government Portals** — IRS, State, e-File
- **Payments** — Stripe, ACH, Cards
- **Calendar** — Appointments
- **Team** — Internal Tasks

A00 AI ORCHESTRATOR / SUPERVISOR AGENT

Routes work between agents, workflows and stages.
Monitors progress, ensures compliance and human oversight.

- Task routing & prioritization
- Cross-agent coordination
- Exception handling
- Real-time monitoring
- Compliance enforcement

CLIENT AGENTS (A01 – A05, A09, A43, A44, A45, A46, A50)

- Client Onboarding (A01)
- Identity Verification (A02)
- Engagement (A03)
- Tax Organizer (A04)
- Document Intake (A05)
- Missing Document (A09)
- Client Communication (A43)
- Appointment (A44)
- Task Management (A45)
- Deadline Tracking (A46)
- Renewal (A50)

and related agents...

TAX AGENTS (A04, A11, A16 – A33, A37, A38, A40 – A42)

- Document Intelligence (A06 – A08)
- Evidence & Validation (A11)
- Bookkeeping & Reconciliation (A12 – A15)
- Book-to-Tax (A16)
- Tax Research (A17)
- Federal / State / Entity / Individual (A18 – A21)
- Deductions, Credits, Depreciation (A22 – A24)
- Payroll & Estimated Tax (A25 – A26)
- Tax Planning (A27)
- Calculation Verification (A28)
- Return Preparation & Mapping (A29 – A31)
- Risk, QC and Readiness (A32 – A37)
- Filing & Government Response (A38 – A42)

and related agents...

OPERATIONS AGENTS (A43 – A50, A58 – A60)

- Workflow & Service Management
- Scheduling (A44)
- Resource Allocation
- Quality Control (A35)
- Billing & Payments (A47 – A48)
- Archive & Renewal (A49 – A50)
- Provider Integrations (A59)
- System Health (A58)
- Practice Intelligence (A60)

and related agents...

ACCOUNTING AGENTS (A12 – A15)

- Bookkeeping (A12)
- Reconciliation (A13)
- Journal Entry (A14)
- Financial Statements (A15)

and related accounting agents...

COMPLIANCE AGENTS (A33, A35, A36, A51 – A53, A56, A57)

- Risk & Materiality (A33)
- Quality Control (A35)
- Signature Readiness (A36)
- Audit Trail (A51)
- Compliance (A52)
- Privacy / Data Protection (A53)
- AI Governance (A56)
- Hallucination Verification (A57)

and related compliance agents...

SECURITY AGENTS (A54 – A55, A58)

- Security Monitoring (A54)
- Access Control (A55)
- System Health (A58)
- Threat Detection
- RBAC/RLS Monitoring
- Anomaly Detection
- Data Protection

and related security agents...

HUMAN AUTHORIZATION LAYER
Critical decisions and final review by authorized personnel.

- **Preparer** — Prepare and review AI work.
- **Reviewer** — Independent review and validation.
- **CPA / EA** — Final approval, sign-off and authorization.

AUTHORITATIVE RECORD
Finalized tax returns, supporting documentation and complete audit trail.

- Immutable audit log
- Version controlled records
- Secure storage (encrypted)
- Long-term retention
- Search and retrieval

INFRASTRUCTURE LAYER
Secure, Scalable and Compliant (e.g., AWS / Azure / GCP)

- **Compute** — Containers / Kubernetes
- **Database** — PostgreSQL (Supabase)
- **File Storage** — Encrypted Storage (S3, GCS)
- **Security** — WAF, Firewall, DLP
- **Backup & DR** — Multi-Region
- **CI/CD** — Automated Deployment

COMPLETE AGENT REGISTRY (A00 – A60) — 61 SPECIALIZED AGENTS

	#	AI Agent	Primary Responsibility	Human Gate
	A00	Supervisor / Orchestrator Agent	Routes work between agents, workflows and stages	Yes (material)
CLIENT AGENTS	A01	Client Onboarding Agent	Guides onboarding and detects missing information	Staff
	A02	Identity Verification Agent	Reviews identity evidence and inconsistencies	Staff
	A03	Engagement Agent	Determines required engagement workflow	Staff
	A04	Tax Organizer Agent	Intelligent taxpaver questionnaire	Preparer
	A05	Document Intake Agent	Controls incoming documents	Staff
	A09	Missing Document Agent	Identifies missing evidence	Staff
	A43	Client Communication Agent	Drafts client communications	Staff
	A44	Appointment Agent	Coordinates appointments	Staff / client
	A45	Task Agent	Creates workflow tasks	Policy-controlled
	A46	Deadline Agent	Tracks filing/payment deadlines	Staff
	A50	Renewal Agent	Prepares next-year engagement	Staff
TAX AGENTS	A06	Document Classification Agent	Classifies W-2, 1099, K-1, receipts, etc.	Reviewer
	A07	OCR Extraction Agent	Extracts structured fields	Reviewer
	A08	Document Validation Agent	Checks extracted info against source	Reviewer
	A11	Evidence Agent	Builds evidence chains	Preparer
	A16	Book-to-Tax Agent	Identifies accounting/tax differences	Tax professional
	A17	Tax Research Agent	Researches applicable tax rules	CPA/EA
	A18	Federal Tax Agent	Analyzes federal tax treatment	Preparer/CPA
	A19	State Tax Agent	Determines state issues	Preparer/CPA
	A20	Entity Tax Agent	LLC, partnership, S-Corp, C-Corp	CPA/EA
	A21	Individual Tax Agent	Individual return analysis	Preparer
	A22	Deduction Agent	Identifies deductions	Preparer
	A23	Credit Agent	Identifies credits	Preparer
	A24	Depreciation Agent	Asset/depreciation treatment	Preparer
	A25	Payroll Tax Agent	Payroll-related analysis	Professional
	A26	Estimated Tax Agent	Projects estimated payments	Professional
	A27	Tax Planning Agent	Scenario and planning analysis	CPA/EA
	A28	Tax Calculation Verification Agent	Checks determinisic outputs	Reviewer
	A29	Return Preparation Agent	Coordinates return assembly	Preparer
	A30	Form Mapping Agent	Maps validated info to tax forms	Preparer
	A31	Return Diagnostics Agent	Detects incomplete/inconsistent returns	Reviewer
	A32	Exception Agent	Creates and routes exceptions	Reviewer
	A33	Risk & Materiality Agent	Scores material risks	CPA/EA
	A37	Filing Readiness Agent	Determines if filing gates satisfied	Authorized filer
	A38	Filing Agent	Coordinates approved submission	Explicit authorization
	A40	Rejection Resolution Agent	Diagnoses rejected filings	Preparer
	A41	Tax Notice Agent	Classifies government notices	CPA/EA
	A42	Amendment Agent	Identifies amendment workflow	CPA/EA
OPERATIONS	A47	Billing Agent	Assists billing/invoice workflows	Authorized staff
	A48	Payment Reconciliation Agent	Matches payments to invoices	Accounting
	A49	Archive Agent	Checks completeness before archival	Staff
	A58	System Health Agent	Monitors services and integration health	Admin
	A59	Provider Integration Agent	Monitors external provider connectivity	Admin
	A60	Practice Intelligence Agent	Operational analytics and workload analysis	Management
ACCOUNTING	A12	Bookkeeping Agent	Suggests transaction classifications	Accountant
	A13	Reconciliation Agent	Identifies reconciliation differences	Accountant
	A14	Journal Entry Agent	Proposes journal entries	Accountant
	A15	Financial Statement Agent	Analyzes P&L, balance sheet, cash flow	Accountant
COMPLIANCE	A34	Reviewer Agent	AI-assisted second-level review	Human reviewer
	A35	Quality-Control Agent	Independent final QC	CPA/EA
	A36	Signature Readiness Agent	Verifies prerequisites before signature	Staff
	A51	Audit Trail Agent	Detects missing/abnormal audit events	Compliance
	A52	Compliance Agent	Evaluates workflow-policy compliance	Compliance officer
	A53	Privacy Agent	Detects sensitive-data handling issues	Security/compliance
	A56	AI Governance Agent	Monitors model use, prompts and AI decisions	Admin/compliance
	A57	Hallucination/Verification Agent	Challenges unsupported AI conclusions	Human review
SECURITY	A54	Security Agent	Monitors suspicious activity	Security/admin
	A55	Access-Control Agent	Reviews RBAC/RLS anomalies	Admin

TaxGuard AI
AI AGENT HIERARCHY

Controlled Orchestration for Secure, Compliant and Accurate Tax Operations

Structured Agent Communication | Human Oversight at Key Gates | Policy-Based Routing | Audit Trail for All Actions

ORCHESTRATION RULES

1. All agents are routed by A00 (Supervisor).
2. Agents communicate only through defined workflows.
3. No direct peer-to-peer communication.
4. Every material action requires human approval.
5. All actions are logged with immutable audit records.
6. Access is controlled by RBAC and data scope.

AGENT CATEGORIES

- Supervisor
- Client Services
- Document Intelligence
- Accounting
- Tax Intelligence
- Risk / QC Control
- Sign / File / Resolve
- Practice Operations
- Governance / Security / Platform

A00 SUPERVISOR AGENT

Orchestrates all agents, routes tasks, enforces policies, monitors progress and ensures compliance.

- Workflow orchestration
- Task routing and prioritization
- Cross-agent coordination
- Real-time monitoring
- Exception handling
- Compliance enforcement

1 CLIENT SERVICES A01 – A04

- A01 Client Onboarding Agent
- A02 Identity Verification Agent
- A03 Engagement Agent
- A04 Tax Organizer Agent

2 DOCUMENT INTELLIGENCE A05 – A11

- A05 Document Intake Agent
- A06 Document Classification Agent
- A07 OCR Extraction Agent
- A08 Document Validation Agent
- A09 Missing Document Agent
- A10 Duplicate/Fraud Indicator Agent
- A11 Evidence Agent

3 ACCOUNTING A12 – A16

- A12 Bookkeeping Agent
- A13 Reconciliation Agent
- A14 Journal Entry Agent
- A15 Financial Statement Agent
- A16 Book-to-Tax Agent

4 TAX INTELLIGENCE A17 – A31

- A17 Tax Research Agent
- A18 Federal Tax Agent
- A19 State Tax Agent
- A20 Entity Tax Agent
- A21 Individual Tax Agent
- A22 Deduction Agent
- A23 Credit Agent
- A24 Depreciation Agent
- A25 Payroll Tax Agent
- A26 Estimated Tax Agent
- A27 Tax Planning Agent
- A28 Tax Calculation Verification Agent
- A29 Return Preparation Agent
- A30 Form Mapping Agent
- A31 Return Diagnostics Agent

5 RISK / QC CONTROL A32 – A35

- A32 Exception Agent
- A33 Risk & Materiality Agent
- A34 Reviewer Agent
- A35 Quality-Control Agent

6 SIGN / FILE / RESOLVE A36 – A42

- A36 Signature Readiness Agent
- A37 Filing Readiness Agent
- A38 Filing Agent
- A39 Government Response Agent
- A40 Rejection Resolution Agent
- A41 Tax Notice Agent
- A42 Amendment Agent

7 PRACTICE OPERATIONS A43 – A50

- A43 Client Communication Agent
- A44 Appointment Agent
- A45 Task Agent
- A46 Deadline Agent
- A47 Billing Agent
- A48 Payment Reconciliation Agent
- A49 Archive Agent
- A50 Renewal Agent

8 GOVERNANCE / SECURITY / PLATFORM A51 – A60

- A51 Audit Trail Agent
- A52 Compliance Agent
- A53 Privacy Agent
- A54 Security Agent
- A55 Access-Control Agent
- A56 AI Governance Agent
- A57 Hallucination/Verification Agent
- A58 System Health Agent
- A59 Provider Integration Agent
- A60 Practice Intelligence Agent

CONTROLLED COMMUNICATION FLOW

A00 SUPERVISOR (Only authorized routing)

- **Tier 1 Agents** — Client Services | Document Intelligence | Accounting (A01 – A16)
- **Tier 2 Agents** — Tax Intelligence (A17 – A31)
- **Tier 3 Agents** — Risk / QC Control (A32 – A35)
- **Tier 4 Agents** — Sign / File / Resolve (A36 – A42)
- **Tier 5 Agents** — Practice Operations (A43 – A50)
- **Tier 6 Agents** — Governance / Security / Platform (A51 – A60)

KEY BENEFITS

- Structured and secure agent hierarchy
- No arbitrary agent communication
- Clear workflow progression
- Human oversight at critical gates
- Complete audit trail
- Scalable and maintainable architecture

TaxGuard AI

AI-POWERED TAX OPERATING SYSTEM

How an Agent Processes Work

Controlled • Secure • Compliant • Auditable

Consistent Process | Human Oversight | Audit Everything

1. **Event / User Request**
   A user, system event or scheduled task starts the process.
2. **A00 Orchestrator**
   Routes the request and manages workflow.
3. **Identify Tax Case**
   Determine client, tax year, entity type and required workflow.
4. **Verify**
   Validate tenant, user, role, permissions and workflow stage.
5. **Select Agent**
   Choose the appropriate specialized agent (A01–A60).
6. **Retrieve Authorized Context**
   Pull only the data the agent is allowed to access.
7. **AI Reasoning**
   Agent analyzes data using rules, models and tools.
8. **Structured Proposal**
   Agent returns a structured and traceable recommendation.
9. **Verification Agent**
   Independent check for accuracy, completeness and compliance.
10. **Confidence + Evidence + Risk**
    Assigns confidence score, shows evidence and risk level.
11. **Decision Path**
    Automatic workflow for low risk; human review for material items.
12. **Human Review (if required)**
    Authorized personnel approve, reject or modify.
13. **Authoritative Action**
    Execute the approved action (e.g., create record, submit return, send communication).
14. **Audit Ledger**
    Immutable record of the entire process.

EVENT / USER REQUEST

User action, scheduled task, or system event (e.g., new document, client request, deadline)

A00 ORCHESTRATOR

Routes request, coordinates agents, manages workflow and enforces policies.

IDENTIFY TAX CASE

Determine client, tax year, entity type, and required workflow stage.

VERIFY

Tenant | User | Role | Permission
Tax Year | Workflow Stage

SELECT AGENT

Choose the appropriate specialized agent (A01 – A60) based on task and context.

RETRIEVE AUTHORIZED CONTEXT

Client Data | Documents | Evidence | Accounting Records | Tax Rules | Prior Decisions

AI REASONING

Analyze data using LLM, deterministic rules, tax knowledge base and tools.

STRUCTURED PROPOSAL

Return structured output with recommendations, form data, explanations and source references.

VERIFICATION AGENT

Independent validation of the agent's output against sources, rules and consistency checks.

CONFIDENCE + EVIDENCE + RISK

Confidence score | Evidence links | Risk assessment

LOW RISK
Meets automated approval criteria.

MATERIAL
Requires human review due to risk, complexity or policy rules.

AUTOMATIC WORKFLOW
Proceed to authorized action based on policy.

HUMAN REVIEW
Preparer → Reviewer → CPA/EA
Approve / Reject / Modiify

AUTHORITATIVE ACTION

Execute approved action
(e.g., create/update record, submit return, send communication, create task, update status).

AUDIT LEDGER

Immutable record of the entire process
(event, inputs, agent output, human actions, final result).

Example Events / Requests

- Client uploads a document
- New client intake
- Prepare tax return
- Answer client question
- Scheduled deadline check
- External notice received

Orchestrator Responsibilities

- Route to correct agents
- Enforce policies and permissions
- Manage workflow and state
- Monitor progress and exceptions
- Ensure compliance and audit logging

Verification Checks

- Valid tenant/organization
- User authentication (MFA)
- Role and permissions (RBAC)
- Client access rights
- Correct tax year
- Allowed workflow stage

Authorized Context (Examples)

- Client profile and tax information
- Uploaded documents (W-2, 1099, etc.)
- Extracted evidence and structured data
- Accounting records (QuickBooks, etc.)
- Applicable tax rules and references
- Prior agent decisions and notes

AI Reasoning

- Use domain knowledge and tax rules
- Analyze documents and data
- Identify issues, opportunities and risks
- Generate structured recommendations
- Cite sources and evidence

Structured Proposal (Example)

```
{
  "recommendation": "Include W-2 income",
  "confidence": 0.92,
  "evidence": ["w2_2024.pdf#p1"],
  "risk": "low",
  "nextSteps": ["update_form", "review"]
}
```

Verification Agent Checks

- Cross-check with source documents
- Validate calculations and logic
- Check for missing information
- Confirm rule applicability
- Flag inconsistencies or hallucinations

Risk Levels

- Low — Automatic workflow
- Medium — Human review (depending on policy)
- Material — Requires human approval

Human Review Actions

- Approve — proceed to authoritative action
- Reject — return to agent with feedback
- Modify — adjust and proceed

Examples of Authoritative Actions

- Create/update client record
- Generate or update tax forms
- Submit return to IRS/state (when authorized)
- Send client communication
- Create tasks and deadlines
- Update workflow stage

Audit Ledger Record (Example)

```
{
  "timestamp": "2026-09-21T14:32:10Z",
  "agent": "A18 Federal Tax Agent",
  "user": "j.doe@firm.com",
  "action": "propose_form_update",
  "result": "approved",
  "evidence": ["w2_2024.pdf#p1"],
  "rish": "low",
  "hash": "a3f5e1...9b7c"
}
```

TaxGuard AI
MULTI-AGENT TAX WORKFLOW

Example Workflow: W-2 Document Processing
Controlled Agents • Human Oversight • Accurate & Auditable

Specialized Agents | Human Approval | Audit Trail for Every Step

1 W-2 UPLOAD
Client uploads W-2 document (Portal or Email).

Input
- PDF, Image, or E-File
- Stored securely



2 A05 – Intake Agent
Controls incoming documents and initiates processing.

A05 Document Intake Agent
- Receives document
- Checks file type and size
- Assigns document ID

3 Security Gate
Validates and scans document for security and compliance.

Security & Compliance
- Virus/malware scan
- PII detection and encryption
- Access control and audit log

Document
Passed Security Gate

4 A06 – Classification Agent
Identifies document type using AI/ML models.

A06 Document Classification Agent
- Analyzes content and structure
- Classifies document type
- Confidence score

"This appears to be Form W-2"
Confidence: 98%

5 A07 – Extraction Agent
Extracts structured data from the document.

A07 OCR Extraction Agent
- Extracts key fields using OCR and AI
- Maps to structured format

Extracted Data (Example)

Employer	ABC Corporation
EIN	12-3456789
Wages	$85,000.00
Federal Withholding	$12,500.00
State Wages	$85,000.00
State Withholding	$4,800.00
etc.	...

6 A08 – Validation Agent
Validates extracted data against source document.

A06 Document Validation Agent
- Checks completeness
- Compares with source
- Validates format and logic

Data Validated
All key fields match source.

7 A10 – Duplicate/Fraud Agent
Checks for duplicates and suspicious indicators.

A10 Duplicate/Fraud Indicator Agent
- Detects duplicates
- Checks for inconsistencies
- Flags suspicious items

No Duplicates Found
No fraud indicators detected.

8 A11 – Evidence Agent
Creates evidence chain for tax positions.

A11 Evidence Agent
- Links source document
- Creates evidence metadata
- Builds audit trace

Evidence Chain Created
- Document ID, hash
- Source, extraction data
- Audit reference

9 HUMAN VALIDATION
Staff reviews extracted data and confirms accuracy.

Staff Reviewer
- Review extracted fields
- Confirm or correct data
- Approve to proceed

Status: Approved
Reviewed by: Staff
Date: 2026-09-21 14:32

10 A18 – Federal Tax Agent
Analyzes federal tax treatment.

A18 Federal Tax Agent
- Applies federal tax rules
- Determines tax treatment
- Identifies considerations

Federal Analysis Complete
- Filing status, withholding
- Applicable adjustments

11 A19 – State Tax Agent
Determines applicable state issues.

A19 State Tax Agent
- Applies state tax rules
- Calculates state impact
- Identifies state-specific items

State Analysis Complete
- Resident state verification
- State withholding validated

12 Deterministic Tax Engine
Performs authoritative tax calculations.

Tax Calculation Engine
- Applies validated data
- Uses deterministic rules
- Generates tax calculations

Tax Calculations Generated
- Federal and state tax
- Line-by-line calculations

13 A28 – Calculation Verification
Independently checks calculation outputs.

A28 Calculation Verification Agent
- Recalculates outputs
- Compares with engine
- Confirms accuracy

Calculations Verified
- Results match (100%)

14 A30 – Form Mapping Agent
Maps validated data to tax forms.

A30 Form Mapping Agent
- Maps data to correct forms
- Populates form fields
- Validates form logic

Forms Mapped
- Form 1040 (with W-2 data)
- All fields populated

15 A31 – Diagnostics Agent
Checks for incomplete or inconsistent returns.

A31 Return Diagnostics Agent
- Detects missing information
- Checks for inconsistencies
- Flags potential issues

Diagnostics Complete
- No critical issues
- Minor suggestions (if any)

16 A34 – Reviewer Agent
Performs AI-assisted second-level review.

A34 Reviewer Agent
- Reviews entire workflow
- Checks compliance and logic
- Prepares review summary

Reviewer Review Complete
- Recommendation: Approve
- No material issues

17 HUMAN REVIEWER
Tax preparer / reviewer performs final review.

Tax Preparer / Reviewer
- Review full return
- Check agent recommendations
- Approve, reject or modify

Approved for Next Stage
Reviewed by: Tax Preparer
Date: 2026-09-21 16:05

AGENTS INVOLVED

5	A05	Document Intake Agent
6	A06	Document Classification Agent
7	A07	OCR Extraction Agent
8	A08	Document Validation Agent
10	A10	Duplicate/Fraud Indicator Agent
11	A11	Evidence Agent
18	A18	Federal Tax Agent
19	A19	State Tax Agent
28	A28	Calculation Verification Agent
30	A30	Form Mapping Agent
31	A31	Return Diagnostics Agent
34	A34	Reviewer Agent

KEY PRINCIPLES

Controlled Orchestration
A00 routes and manages the workflow.

Specialized Agents
Each agent has a defined role and scope.

No Arbitrary Communication
Agents communicate only through controlled workflows.

Human Oversight
Key decisions require human approval.

Immutable Audit Trail
Every step is logged with evidence.

Secure and Compliant
Designed for U.S. tax and data security requirements.

DATA FLOW (SIMPLIFIED)

Input Document
(Stored Securely)
↓
Agent Processing
(Each Agent Has Limited Access)
↓
Structured Data
(Validated and Approved)
↓
Tax Engine
(Deterministic Calculations)
↓
Form Generation
(Validated and Reviewed)
↓
Human Approval
(Preparer → Reviewer)
↓
Final Tax Return
(Authorized for Filing)
↓
Audit Ledger
(Immutable Record)

RESULT

- Accurate and validated W-2 data
- Proper tax treatment (federal & state)
- Mapped to correct tax forms
- Reviewed by AI and humans
- Complete audit trail
- Ready for next stage (e.g., full return, filing or client review)

TaxGuard AI

TAX RETURN MULTI-AGENT PIPELINE

From Data to Approved Tax Return

AI Agents • Deterministic Calculations • Human Oversight

Accurate | Compliant | Secure | Auditable

INPUT SOURCES

1. **CLIENT DATA**
   - Client profile
   - Tax questionnaire
   - Filing status
   - Prior year data
2. **DOCUMENTS**
   - W-2, 1099, K-1, etc.
   - Receipts and statements
   - PDF, image, e-file
   - OCR extracted data
3. **ACCOUNTING**
   - General ledger
   - Profit & Loss
   - Balance sheet
   - Trial balance
   - Bank and credit card feeds

AGENT GROUPS

Evidence & Data Agents

- A11 Evidence Agent — Builds verified evidence chain
- A16 Book-to-Tax Agent — Identifies accounting/tax differences

Research & Analysis Agents

- A17 Tax Research Agent — Researches applicable tax rules

Tax Domain Agents

- A18 Federal Tax Agent
- A19 State Tax Agent
- A20 Entity Tax Agent
- A21 Individual Tax Agent
- A22–A27 Specialized Agents — Deduction, Credit, Depreciation, Payroll, Estimated Tax, Tax Planning

Calculation & Preparation Agents

- A28 Calculation Verification Agent
- A29 Return Preparation Agent
- A30 Form Mapping Agent
- A31 Diagnostics Agent

Risk & Review Agents

- A33 Risk & Materiality Agent
- A34 Reviewer Agent
- A35 Independent QC Agent

Human Oversight

- CPA / EA — Final review and approval

CLIENT DATA

- Profile & Questionnaire
- Filing Status
- Prior Year Data

DOCUMENTS

- W-2, 1099, K-1, etc.
- Statements & Receipts
- OCR Extracted Data

ACCOUNTING

- General Ledger
- P&L, Balance Sheet
- Bank & Card Feeds

1. **A11 Evidence Agent** — Builds verified evidence chain from all sources.
   - Link documents to data
   - Create source references
   - Build audit evidence
2. **A16 Book-to-Tax Agent** — Identifies accounting/tax differences.
   - Compare book vs tax
   - Identify adjustments
   - Flag reconciling items
3. **A17 Tax Research Agent** — Researches applicable tax rules.
   - Federal, state and local rules
   - Entity and individual rules
   - Apply to client situation
4. **TAX DOMAIN AGENTS (PARALLEL ANALYSIS)**
   - A18 Federal Tax Agent
   - A19 State Tax Agent
   - A20 Entity Tax Agent
   - A21 Individual Tax Agent
   - A22 – A27 Specialized Agents: Deduction, Credit, Depreciation, Payroll, Estimated Tax, Tax Planning
5. **TAX RULE ENGINE** — Applies validated tax rules, forms, limits and logic.
   - Federal & state tax rules
   - Credits and deductions
   - Entity and individual rules
   - Updated tax law library
6. **DETERMINISTIC CALCULATION** — Computes tax using validated logic and data.
   - Reproducible calculations
   - Supports all entity types
   - Applies credits, deductions
   - Generates tax results
7. **A28 Calculation Verification Agent** — Independently checks calculation outputs.
   - Recalculate tax amounts
   - Compare with engine results
   - Verify accuracy and limits
8. **A29 Return Preparation Agent** — Coordinates return assembly.
   - Compile validated data
   - Prepare return data sets
   - Coordinate multi-state returns
9. **A30 Form Mapping Agent** — Maps data to tax forms.
   - Map to correct IRS/state forms
   - Populate form fields
   - Validate form logic and requirements
10. **A31 Diagnostics Agent** — Checks for incomplete or inconsistent returns.
    - Detect missing information
    - Check for inconsistencies
    - Flag potential issues
11. **A33 Risk & Materiality Agent** — Scores material tax risks.
    - Identify high-risk positions
    - Materiality assessment
    - Flag items for review
12. **A34 Reviewer Agent** — Performs AI-assisted second-level review.
    - Review entire return
    - Check compliance and logic
    - Prepare review summary
13. **A35 Independent QC Agent** — Final quality control before human review.
    - Independent validation
    - Ensure all requirements met
    - Confirm readiness for signature
14. **HUMAN CPA / EA** — Final professional review.
    - Review return and workpapers
    - Address flagged items
    - Approve, reject or request changes
15. **APPROVAL** — Authorize filing or finalization.
    - Sign-off by CPA/EA
    - Approve for e-filing or delivery
    - Finalize return and archive

KEY CONTROLS

- **Controlled Orchestration** — Agents work in sequence. No arbitrary communication.
- **Role-Based Access** — Each agent has defined permissions and data scope.
- **Human Gates** — Material actions require human approval.
- **Audit Trail** — Every step is logged with evidence and reasoning.
- **Deterministic Calculations** — Tax amounts calculated using validated rules and logic.
- **Compliance by Design** — Built for IRS and state compliance requirements.

DATA FLOW

- **Client Data** — Profile, questionnaire, prior year data
- **Documents** — W-2, 1099, K-1, receipts, bank statements, etc.
- **Accounting** — GL, P&L, balance sheet, bank and card feeds

↓

Evidence & Analysis (A11 – A17)

↓

Tax Domain Analysis (A18 – A27)

↓

Rule Engine & Calculation

↓

Preparation & Mapping (A28 – A31)

↓

Risk & Review (A33 – A35)

↓

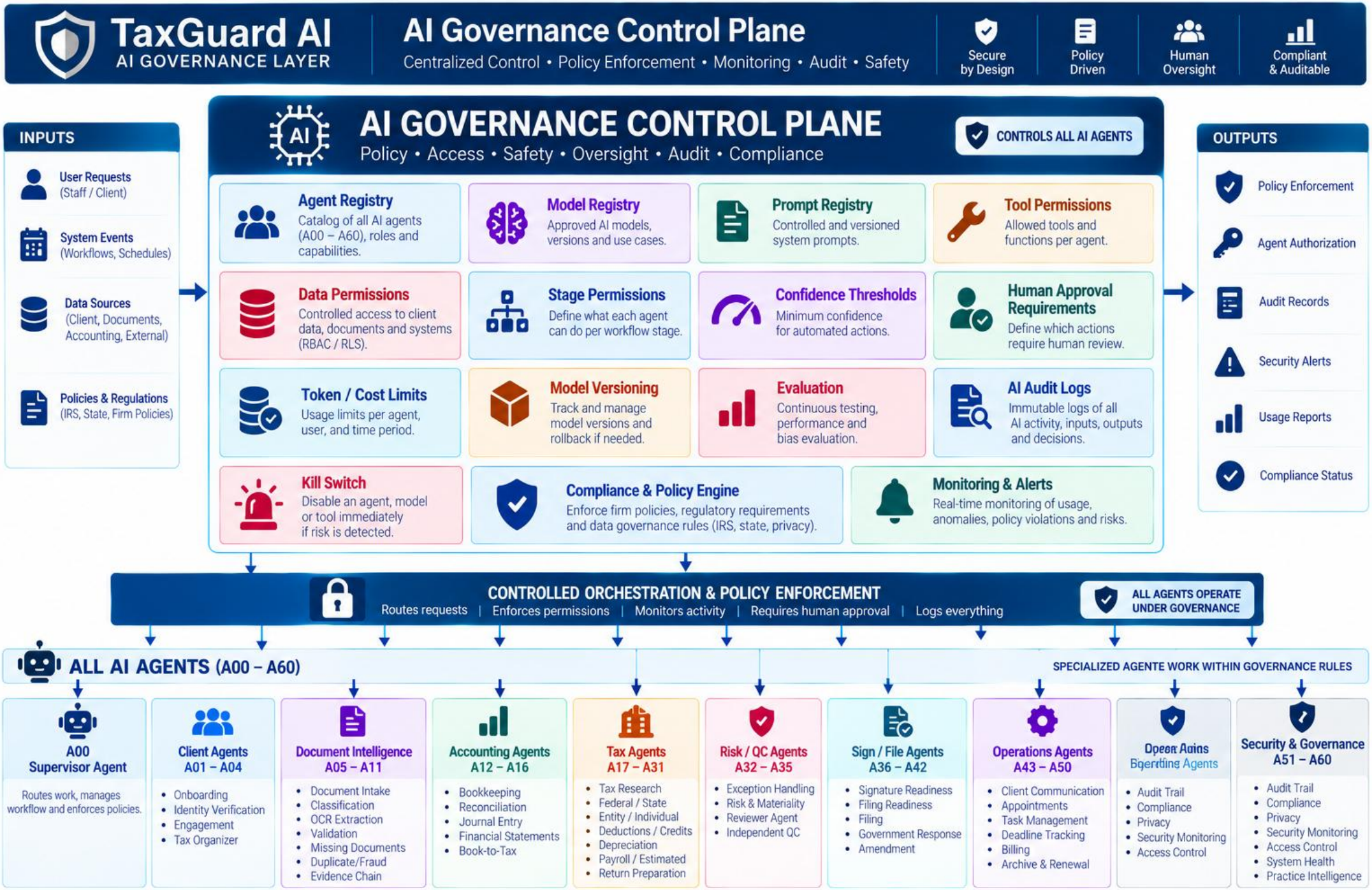
Human Approval (CPA / EA)

↓

Final Tax Return (Approved for Filing)

FINAL OUTPUTS

- **Completed Tax Return** — PDF and e-file ready
- **Workpapers & Evidence** — Linked source documents
- **Audit Record** — Immutable audit trail
- **Filing Status** — Submission and confirmations


TaxGuard AI
AI GOVERNANCE LAYER
AI Governance Control Plane
Centralized Control • Policy Enforcement • Monitoring • Audit • Safety
Secure by Design
Policy Driven
Human Oversight
Compliant & Auditable
INPUTS
User Requests (Staff / Client)
System Events (Workflows, Schedules)
Data Sources (Client, Documents, Accounting, External)
Policies & Regulations (IRS, State, Firm Policies)
AI GOVERNANCE CONTROL PLANE
Policy • Access • Safety • Oversight • Audit • Compliance
CONTROLS ALL AI AGENTS
Agent Registry
Catalog of all AI agents (A00 – A60), roles and capabilities.
Model Registry
Approved AI models, versions and use cases.
Prompt Registry
Controlled and versioned system prompts.
Tool Permissions
Allowed tools and functions per agent.
Data Permissions
Controlled access to client data, documents and systems (RBAC / RLS).
Stage Permissions
Define what each agent can do per workflow stage.
Confidence Thresholds
Minimum confidence for automated actions.
Human Approval Requirements
Define which actions require human review.
Token / Cost Limits
Usage limits per agent, user, and time period.
Model Versioning
Track and manage model versions and rollback if needed.
Evaluation
Continuous testing, performance and bias evaluation.
AI Audit Logs
Immutable logs of all AI activity, inputs, outputs and decisions.
Kill Switch
Disable an agent, model or tool immediately if risk is detected.
Compliance & Policy Engine
Enforce firm policies, regulatory requirements and data governance rules (IRS, state, privacy).
Monitoring & Alerts
Real-time monitoring of usage, anomalies, policy violations and risks.
OUTPUTS
Policy Enforcement
Agent Authorization
Audit Records
Security Alerts
Usage Reports
Compliance Status
CONTROLLED ORCHESTRATION & POLICY ENFORCEMENT
Routes requests | Enforces permissions | Monitors activity | Requires human approval | Logs everything
ALL AI AGENTS OPERATE UNDER GOVERNANCE
ALL AI AGENTS (A00 – A60)
SPECIALIZED AGENTE WORK WITHIN GOVERNANCE RULES
A00 Supervisor Agent
Routes work, manages workflow and enforces policies.
Client Agents A01 – A04
Onboarding
Identity Verification
Engagement
Tax Organizer
Document Intelligence A05 – A11
Document Intake
Classification
OCR Extraction
Validation
Missing Documents
Duplicate/Fraud
Evidence Chain
Accounting Agents A12 – A16
Bookkeeping
Reconciliation
Journal Entry
Financial Statements
Book-to-Tax
Tax Agents A17 – A31
Tax Research
Federal / State
Entity / Individual
Deductions / Credits
Depreciation
Payroll / Estimated
Return Preparation
Risk / QC Agents A32 – A35
Exception Handling
Risk & Materiality
Reviewer Agent
Independent QC
Sign / File Agents A36 – A42
Signature Readiness
Filing Readiness
Filing
Government Response
Amendment
Operations Agents A43 – A50
Client Communication
Appointments
Task Management
Deadline Tracking
Billing
Archive & Renewal
Security & Governance A51 – A60
Audit Trail
Compliance
Privacy
Security Monitoring
Access Control
System Health
Practice Intelligence

TaxGuard AI
AI RUN AUDIT RECORD

Every AI Execution Creates an ai_run Record
Complete traceability, accountability and reproducibility for every AI-assisted decision.

Traceable Decisions | Audit Ready Compliance | Immutable Records | Reconstruct Any Decision

1. AI EXECUTION FLOW

User Request / Event
(e.g., upload document, run analysis)

↓

A00 Orchestrator
Selects agent and context

↓

Selected AI Agent
(A01 – A60)
Processes request

↓

AI Execution
Uses model, prompt and tools

↓

Generate Output
Structured proposal, result or recommendation

↓

Create ai_run Record
Log all details of the execution

↓

Human Review (if required)
Approve / Reject / Modify

↓

Authoritative Action
Update case, create documents, etc.

2. SAMPLE ai_run RECORD

JSON

```json
{
  "agent_id": "A07",
  "agent_version": "1.2.0",
  "model_provider": "OpenAI",
  "model_version": "gpt-4o-2024-08-06",
  "prompt_version": "v3.1",
  "tenant_id": "TEN-001",
  "client_id": "CLT-2026-000123",
  "tax_case_id": "TAX-2026-000456",
  "tax_year": 2024,
  "workflow_stage": "M2-Document-Intelligence",
  "requested_action": "extract_w2_fields",
  "source_evidence": [
    {
      "document_id": "DOC-001",
      "document_name": "W2_ABC_Corp.pdf",
      "hash": "a3f5e1...9b7c",
      "page": "1"
    }
  ],
  "input_hash": "d41d8cd98f00b204e9800998ecf8427e",
  "output_hash": "7e9c2d1a4b6f3e8c91d2a5b7c6d4e1f0",
  "confidence": 0.92,
  "risk_level": "low",
  "proposed_action": "store_extracted_w2_data",
  "human_review_required": true,
  "reviewer": "USR-2026-0045",
  "review_decision": "approved",
  "timestamp": "2026-09-21T14:32:18Z",
  "audit_event_id": "AUD-2026-09-21-000789"
}
```

3. ai_run FIELDS DESCRIPTION

Field	Type	Description
agent_id	string	ID of the AI agent that executed (A00–A60).
agent_version	string	Version of the agent logic.
model_provider	string	LLM provider (e.g., OpenAI, Anthropic, Google).
model_version	string	Specific model version used.
prompt_version	string	Version of the system prompt/template.
tenant_id	string	Organization/tenant ID.
client_id	string	Client ID (e.g., AR-CLT-YYYY-XXXXX).
tax_case_id	string	Tax case/engagement ID.
tax_year	number	Tax year being processed.
workflow_stage	string	Current workflow stage (e.g., M2 Collect, M3 Validate).
requested_action	string	Action requested (e.g., extract_w2_fields, classify_document).
source_evidence[]	array	List of source documents, IDs, hashes, pages, etc.
input_hash	string	Hash of the input data/context sent to the model.
output_hash	string	Hash of the model output/response.
confidence	number	Agent's confidence score (0–1).
risk_level	string	Risk level (low, medium, high, material).
proposed_action	string	Recommended next action.
human_review_required	boolean	Whether human review is required.
reviewer	string	User ID of reviewer (if applicable).
review_decision	string	Approved, rejected, modified, or pending.
timestamp	datetime	When the execution occurred (UTC).
audit_event_id	string	Unique audit event ID linking to full audit log entry.

4. STORE IN AUDIT DATABASE

ai_run Table

- Immutable log record
- Linked to audit ledger
- Secure and encrypted
- Indexed for search and reporting

5. LINKED RECORDS

- Client Record
- Document Records
- Tax Case Record
- Audit Ledger Entry

6. AUDIT TRAIL

- ✔ Complete execution history
- ✔ Input and output hashes
- ✔ Model and prompt versions
- ✔ Human review actions
- ✔ Reconstruct exact context
- ✔ Regulatory and internal compliance

7. USE CASES

- ✔ Reconstruct how a decision was reached
- ✔ Investigate discrepancies
- ✔ Respond to client or regulatory inquiries
- ✔ Monitor agent performance
- ✔ Detect unusual activity
- ✔ Support audits and quality control

TaxGuard AI
AGENT PERMISSIONS & CONTROLS

AI Agents Operate with Defined Permissions
Each agent has specific capabilities and restrictions, enforced by the governance control plane.

Policy Enforced | Role-Based Access | Stage Controlled | Audit Logged

PERMISSION MODEL

Role-Based Permissions
Each agent has defined read, write and action permissions.

Stage-Based Access
Agents can only act within allowed workflow stages.

Data Scope Control
Limited to authorized tenant, client, tax year and case.

Tool Permissions
Only approved tools and integrations per agent.

Policy Enforcement
Prevent unauthorized actions (e.g., no filing, no approval).

Audit & Monitoring
All actions are logged and reviewed.

EXAMPLE 1: A07 OCR EXTRACTION AGENT

A07
OCR EXTRACTION AGENT
Extracts structured data from documents using OCR and AI.

Primary Responsibilities
- Read authorized documents
- Extract structured fields (e.g., W-2, 1099, K-1, invoices)
- Provide confidence scores
- Create AI proposal for reviewer

CAN (Allowed Permissions)
- ✓ Read authorized document
- ✓ Extract fields using OCR/AI
- ✓ Provide confidence score
- ✓ Create structured proposal
- ✓ Store extracted data in staging
- ✓ Link extracted data to document
- ✓ Create evidence metadata
- ✓ Log activity to audit trail

CANNOT (Restricted Permissions)
- ✗ Mark document tax-verified
- ✗ Modify authoritative return data
- ✗ Approve tax position
- ✗ File return
- ✗ Sign return
- ✗ Access unrelated client data
- ✗ Delete source documents

EXAMPLE 2: A38 FILING AGENT

A38
FILING AGENT
Coordinates approved provider submission and records response.

Primary Responsibilities
- Inspect approved filing package
- Validate filing prerequisites
- Communicate with approved provider (e.g., IRS, state portals)
- Record provider response (acknowledgments, rejections)
- Update case status

CAN (Allowed Permissions)
- ✓ Inspect approved filing package
- ✓ Validate filing prerequisites
- ✓ Communicate with approved provider
- ✓ Record provider response
- ✓ Update filing status and case records
- ✓ Log all activity to audit trail
- ✓ Access only authorized client and tax year
- ✓ Use approved filing integrations

CANNOT (Restricted Permissions)
- ✗ Bypass approval
- ✗ Create client signature
- ✗ Alter final return
- ✗ Fabricate government acceptance
- ✗ Submit without required authorization
- ✗ Modify approved return data
- ✗ Access or file for unauthorized clients

AGENT PERMISSIONS MATRIX (EXAMPLE)

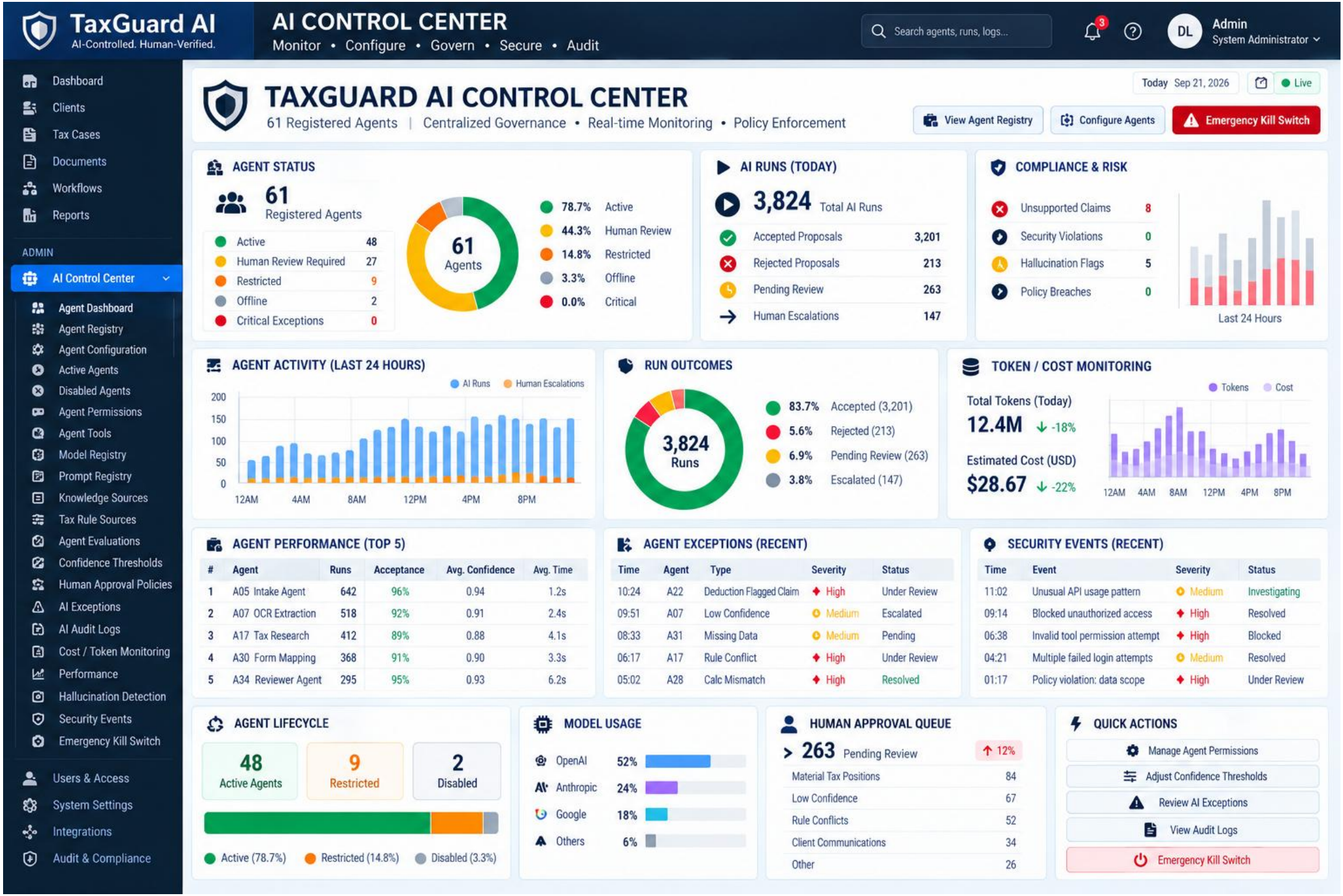
Agent ID	Agent Name	Data Access						Actions						Tools & Integrations				Workflow Stages				
		Read Client Data	Read Documents	Read Accounting	Read Tax Rules	Write Staging	Write Return Data	Create Proposal	Mark Verified	Approve Tax Position	Modify Return	File Return	Sign Return	OCR	LLM	Tax Engine	Filing Provider	Collect	Validate	Prepare	Review	File
A05	Document Intake Agent	✓	✓	✗	✓	✓	✗	✓	✗	✗	✗	✗	✗	○	✗	✗	✗	✓	✓	✗	✗	✗
A06	Classification Agent	✓	✓	✗	✓	✓	✗	✓	✗	✗	✗	✗	✗	○	✗	✗	✗	✓	✓	✗	✗	✗
A07	OCR Extraction Agent	✓	✓	○	○	○	✗	✓	✗	✗	✗	✗	✗	✓	✓	✗	✗	✓	✓	✗	✗	✗
A08	Document Validation Agent	✓	✓	✓	✓	✓	✗	✓	✗	✗	✗	✗	✗	○	✓	✗	✗	✓	✓	✗	✗	✗
A11	Evidence Agent	✓	✓	✓	✓	✓	✗	✓	✗	✗	✗	✗	✗	○	✗	✗	✗	✓	✓	✓	✗	✗
A16	Book-to-Tax Agent	✓	✓	✓	✓	✓	✓	✓	✗	○	○	✗	✗	○	✗	✗	✗	✗	✓	✓	✓	✗
A18	Federal Tax Agent	✓	✓	✗	✓	✓	✓	✓	✗	✗	○	✗	✗	○	✗	✗	✗	✗	✓	✓	✓	✗
A19	State Tax Agent	✓	✓	✗	✓	✓	✓	✓	✗	✗	○	✗	✗	○	✗	✗	✗	✗	✓	✓	✓	✗
A20	Entity Tax Agent	✓	✓	✗	✓	✓	✓	✓	✗	✗	○	✗	✗	○	✗	✗	✗	✗	✓	✓	✓	✗
A29	Return Preparation Agent	✓	✓	✗	○	✓	✓	✓	✓	○	○	✗	✗	○	✗	✗	✗	○	✗	✗	✗	✗
A34	Reviewer Agent	✓	✓	✓	✓	✓	✓	✓	✗	○	✓	✗	✗	○	✗	✗	✗	✓	✗	✗	✗	✗
A35	Independent QC Agent	✓	✓	✓	✓	✓	✓	✓	✗	○	✓	✗	✗	○	✗	✗	✗	✓	✓	✗	✗	✗
A38	Filing Agent	✓	✓	○	○	○	○	○	✗	○	✗	✗	✗	✓	○	✗	✓	✓	✓	✓	✓	✓

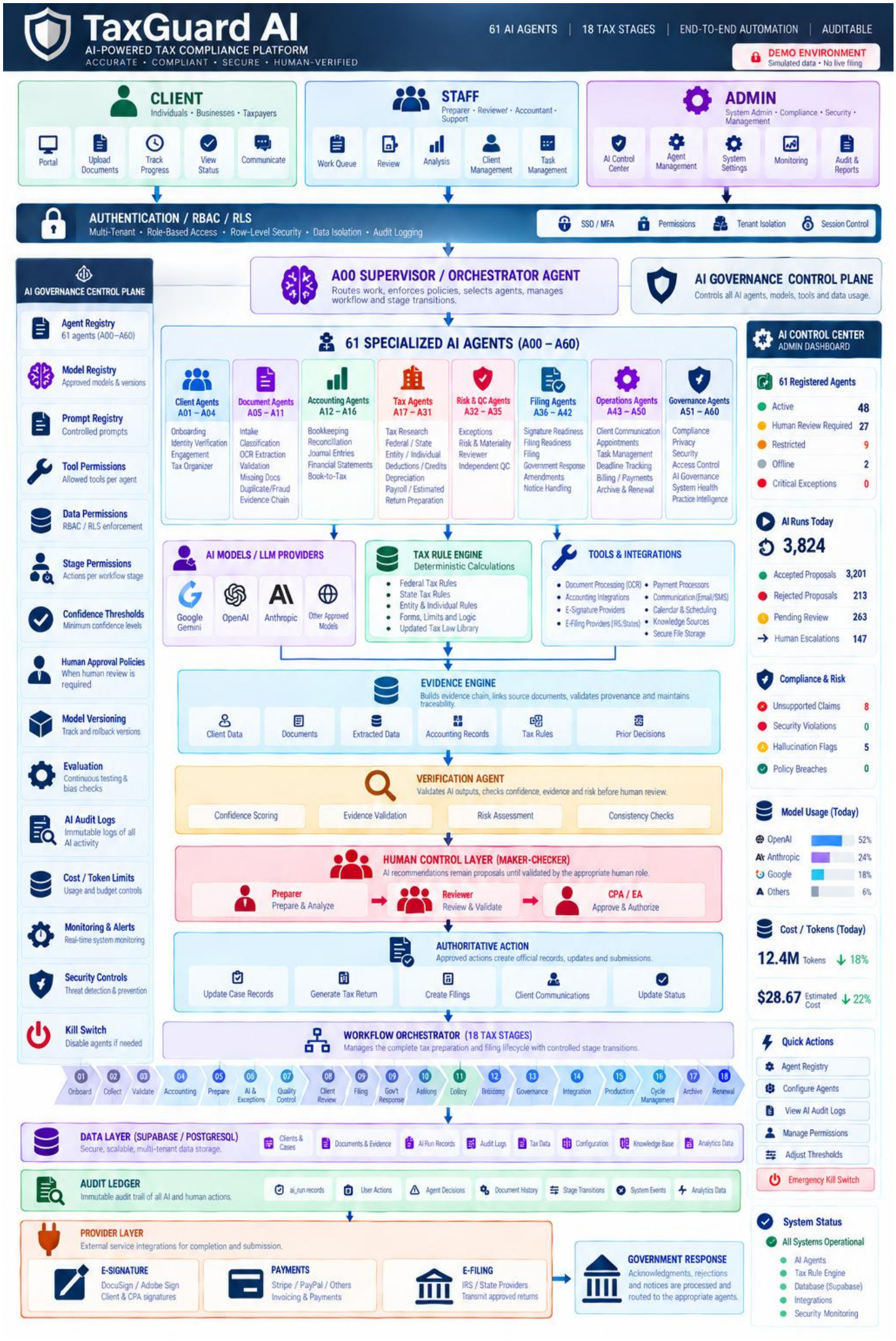
Permission Legend
✓ Allowed | ✗ Not Allowed | ○ Conditional (Policy Controlled)

Controlled by Governance Layer
- Agent roles and scopes
- Tool and integration access
- Data permissions (RBAC / RLS)
- Stage permissions
- Confidence thresholds
- Human approval requirements
- Audit logging and monitoring

Permission Examples
- A07 can extract data but cannot mark as tax-verified.
- A38 can file with provider but cannot bypass approval.
- Only human reviewers and CPA/EA can approve tax positions.
- No AI agent can sign a return or create a client signature.

Enforced at Runtime
- Every action checked against permissions.
- Unauthorized actions are blocked.
- All attempts are logged in the audit trail.
- Kill switch available for any agent.


TaxGuard AI
AI-Controlled. Human-Verified.
AI CONTROL CENTER
Monitor • Configure • Govern • Secure • Audit
Search agents, runs, logs...
Admin
System Administrator
Dashboard
Clients
Tax Cases
Documents
Workflows
Reports
ADMIN
AI Control Center
Agent Dashboard
Agent Registry
Agent Configuration
Active Agents
Disabled Agents
Agent Permissions
Agent Tools
Model Registry
Prompt Registry
Knowledge Sources
Tax Rule Sources
Agent Evaluations
Confidence Thresholds
Human Approval Policies
AI Exceptions
AI Audit Logs
Cost / Token Monitoring
Performance
Hallucination Detection
Security Events
Emergency Kill Switch
Users & Access
System Settings
Integrations
Audit & Compliance
TAXGUARD AI CONTROL CENTER
61 Registered Agents | Centralized Governance • Real-time Monitoring • Policy Enforcement
Today Sep 21, 2026
Live
View Agent Registry
Configure Agents
Emergency Kill Switch
AGENT STATUS
61
Registered Agents
Active 48
Human Review Required 27
Restricted 9
Offline 2
Critical Exceptions 0
61
Agents
78.7% Active
44.3% Human Review
14.8% Restricted
3.3% Offline
0.0% Critical
AI RUNS (TODAY)
3,824 Total AI Runs
Accepted Proposals 3,201
Rejected Proposals 213
Pending Review 263
Human Escalations 147
COMPLIANCE & RISK
Unsupported Claims 8
Security Violations 0
Hallucination Flags 5
Policy Breaches 0
Last 24 Hours
AGENT ACTIVITY (LAST 24 HOURS)
AI Runs
Human Escalations
200
150
100
50
0
12AM
4AM
8AM
12PM
4PM
8PM
RUN OUTCOMES
3,824
Runs
83.7% Accepted (3,201)
5.6% Rejected (213)
6.9% Pending Review (263)
3.8% Escalated (147)
TOKEN / COST MONITORING
Total Tokens (Today)
12.4M ↓ -18%
Estimated Cost (USD)
$28.67 ↓ -22%
Tokens
Cost
12AM
4AM
8AM
12PM
4PM
8PM
AGENT PERFORMANCE (TOP 5)
Agent Runs Acceptance Avg. Confidence Avg. Time
1 A05 Intake Agent 642 96% 0.94 1.2s
2 A07 OCR Extraction 518 92% 0.91 2.4s
3 A17 Tax Research 412 89% 0.88 4.1s
4 A30 Form Mapping 368 91% 0.90 3.3s
5 A34 Reviewer Agent 295 95% 0.93 6.2s
AGENT EXCEPTIONS (RECENT)
Time Agent Type Severity Status
10:24 A22 Deduction Flagged Claim High Under Review
09:51 A07 Low Confidence Medium Escalated
08:33 A31 Missing Data Medium Pending
06:17 A17 Rule Conflict High Under Review
05:02 A28 Calc Mismatch High Resolved
SECURITY EVENTS (RECENT)
Time Event Severity Status
11:02 Unusual API usage pattern Medium Investigating
09:14 Blocked unauthorized access High Resolved
06:38 Invalid tool permission attempt High Blocked
04:21 Multiple failed login attempts Medium Resolved
01:17 Policy violation: data scope High Under Review
AGENT LIFECYCLE
48
Active Agents
9
Restricted
2
Disabled
Active (78.7%)
Restricted (14.8%)
Disabled (3.3%)
MODEL USAGE
OpenAI 52%
Anthropic 24%
Google 18%
Others 6%
HUMAN APPROVAL QUEUE
> 263 Pending Review
↑ 12%
Material Tax Positions 84
Low Confidence 67
Rule Conflicts 52
Client Communications 34
Other 26
QUICK ACTIONS
Manage Agent Permissions
Adjust Confidence Thresholds
Review AI Exceptions
View Audit Logs
Emergency Kill Switch


TaxGuard AI
AI-POWERED TAX COMPLIANCE PLATFORM
ACCURATE • COMPLIANT • SECURE • HUMAN-VERIFIED
61 AI AGENTS | 18 TAX STAGES | END-TO-END AUTOMATION | AUDITABLE
DEMO ENVIRONMENT
Simulated data • No live filing
CLIENT
Individuals • Businesses • Taxpayers
Portal
Upload Documents
Track Progress
View Status
Communicate
STAFF
Preparer • Reviewer • Accountant • Support
Work Queue
Review
Analysis
Client Management
Task Management
ADMIN
System Admin • Compliance • Security • Management
AI Control Center
Agent Management
System Settings
Monitoring
Audit & Reports
AUTHENTICATION / RBAC / RLS
Multi-Tenant • Role-Based Access • Row-Level Security • Data Isolation • Audit Logging
SSO / MFA
Permissions
Tenant Isolation
Session Control
AI GOVERNANCE CONTROL PLANE
Agent Registry
61 agents (A00–A60)
Model Registry
Approved models & versions
Prompt Registry
Controlled prompts
Tool Permissions
Allowed tools per agent
Data Permissions
RBAC / RLS enforcement
Stage Permissions
Actions per workflow stage
Confidence Thresholds
Minimum confidence levels
Human Approval Policies
When human review is required
Model Versioning
Track and rollback versions
Evaluation
Continuous testing & bias checks
AI Audit Logs
Immutable logs of all AI activity
Cost / Token Limits
Usage and budget controls
Monitoring & Alerts
Real-time system monitoring
Security Controls
Threat detection & prevention
Kill Switch
Disable agents if needed
A00 SUPERVISOR / ORCHESTRATOR AGENT
Routes work, enforces policies, selects agents, manages workflow and stage transitions.
AI GOVERNANCE CONTROL PLANE
Controls all AI agents, models, tools and data usage.
61 SPECIALIZED AI AGENTS (A00 – A60)
Client Agents A01 – A04
Onboarding
Identity Verification
Engagement
Tax Organizer
Document Agents A05 – A11
Intake
Classification
OCR Extraction
Validation
Missing Docs
Duplicate/Fraud
Evidence Chain
Accounting Agents A12 – A16
Bookkeeping
Reconciliation
Journal Entries
Financial Statements
Book-to-Tax
Tax Agents A17 – A31
Tax Research
Federal / State
Entity / Individual
Deductions / Credits
Depreciation
Payroll / Estimated
Return Preparation
Risk & QC Agents A32 – A35
Exceptions
Risk & Materiality
Reviewer
Independent QC
Filing Agents A36 – A42
Signature Readiness
Filing Readiness
Filing
Government Response
Amendments
Notice Handling
Operations Agents A43 – A50
Client Communication
Appointments
Task Management
Deadline Tracking
Billing / Payments
Archive & Renewal
Governance Agents A51 – A60
Compliance
Privacy
Security
Access Control
AI Governance
System Health
Practice Intelligence
AI MODELS / LLM PROVIDERS
Google Gemini
OpenAI
Anthropic
Other Approved Models
TAX RULE ENGINE
Deterministic Calculations
Federal Tax Rules
State Tax Rules
Entity & Individual Rules
Forms, Limits and Logic
Updated Tax Law Library
TOOLS & INTEGRATIONS
Document Processing (OCR)
Accounting Integrations
E-Signature Providers
E-Filing Providers (IRS, States)
Payment Processors
Communication (Email/SMS)
Calendar & Scheduling
Knowledge Sources
Secure File Storage
EVIDENCE ENGINE
Builds evidence chain, links source documents, validates provenance and maintains traceability.
Client Data
Documents
Extracted Data
Accounting Records
Tax Rules
Prior Decisions
VERIFICATION AGENT
Validates AI outputs, checks confidence, evidence and risk before human review.
Confidence Scoring
Evidence Validation
Risk Assessment
Consistency Checks
HUMAN CONTROL LAYER (MAKER-CHECKER)
AI recommendations remain proposals until validated by the appropriate human role.
Preparer
Prepare & Analyze
Reviewer
Review & Validate
CPA / EA
Approve & Authorize
AUTHORITATIVE ACTION
Approved actions create official records, updates and submissions.
Update Case Records
Generate Tax Return
Create Filings
Client Communications
Update Status
WORKFLOW ORCHESTRATOR (18 TAX STAGES)
Manages the complete tax preparation and filing lifecycle with controlled stage transitions.
01 Onboard
02 Collect
03 Validate
04 Accounting
05 Prepare
06 AI & Exceptions
07 Quality Control
08 Client Review
09 Filing
09 Gov't Response
10 Actions
11 Entity
12 Billing
13 Governance
14 Integration
15 Production
16 Cycle Management
17 Archive
18 Renewal
DATA LAYER (SUPABASE / POSTGRESQL)
Secure, scalable, multi-tenant data storage.
Clients & Cases
Documents & Evidence
AI Run Records
Audit Logs
Tax Data
Configuration
Knowledge Base
Analytics Data
AUDIT LEDGER
Immutable audit trail of all AI and human actions.
ai_run records
User Actions
Agent Decisions
Document History
Stage Transitions
System Events
Analytics Data
PROVIDER LAYER
External service integrations for completion and submission.
E-SIGNATURE
DocuSign / Adobe Sign
Client & CPA signatures
PAYMENTS
Stripe / PayPal / Others
Invoicing & Payments
E-FILING
IRS / State Providers
Transmit approved returns
GOVERNMENT RESPONSE
Acknowledgments, rejections and notices are processed and routed to the appropriate agents.
AI CONTROL CENTER
ADMIN DASHBOARD
61 Registered Agents
Active 48
Human Review Required 27
Restricted 9
Offline 2
Critical Exceptions 0
AI Runs Today
3,824
Accepted Proposals 3,201
Rejected Proposals 213
Pending Review 263
Human Escalations 147
Compliance & Risk
Unsupported Claims 8
Security Violations 0
Hallucination Flags 5
Policy Breaches 0
Model Usage (Today)
OpenAI 52%
Anthropic 24%
Google 18%
Others 6%
Cost / Tokens (Today)
12.4M Tokens ↓ 18%
$28.67 Estimated Cost ↓ 22%
Quick Actions
Agent Registry
Configure Agents
View AI Audit Logs
Manage Permissions
Adjust Thresholds
Emergency Kill Switch
System Status
All Systems Operational
AI Agents
Tax Rule Engine
Database (Supabase)
Integrations
Security Monitoring